

Logistic regression

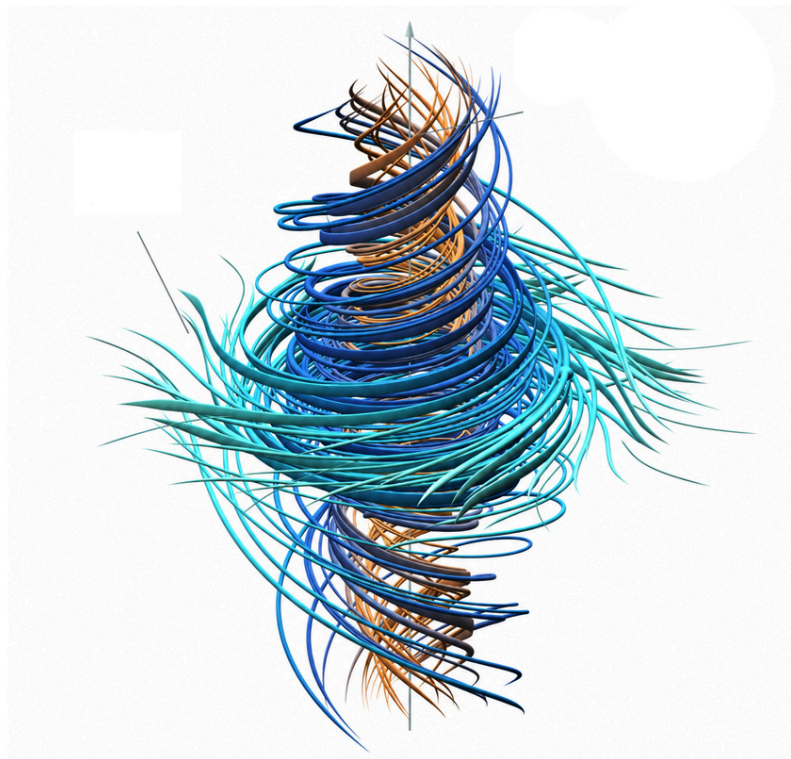
CSI 4106 - Fall 2026

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Preamble

Message of the Day



["OpenAI breakthrough triggers 'existential crisis' in math"](#), C. Zhao, *Science*, 2026-09-15.

On September 8, OpenAI announced that an internal AI system had produced a solution to the Navier–Stokes existence and smoothness problem, one of the Millennium Prize Problems. OpenAI released both a written proof and a formalization in Lean. The mathematical community is still examining the result, so describe it as a **claimed solution**, not as a settled achievement (Zhao 2026).

Here, "existential crisis" refers to a debate about the purpose and future of mathematics, not an existential threat to humanity. The article asks what mathematical understanding

means when an AI system produces a formally verified proof that humans struggle to understand. It also raises concerns about attribution and unequal access to the computing resources required for this kind of research.

- Does a formally verified proof provide mathematical understanding, or only mathematical certainty?
- How might access to powerful AI systems change who receives opportunities and recognition in research?

Today, we return from this frontier system to one of the simplest and most interpretable classification models.

Image credit: OpenAI, via *Science*.

Learning Objectives

- **Distinguish** binary, multiclass, and multilabel classification tasks.
- **Explain** how logistic regression converts a linear score into a probability using the sigmoid function.
- **Relate** the 0.5 classification threshold to the linear decision boundary.
- **Explain** how one-vs-rest extends a binary classifier to multiclass problems.
- **Train** a logistic regression model without leaking information from the test set.
- **Interpret** the coefficients learned by one-vs-rest digit classifiers.

Classification tasks

Definitions

- **Binary classification** is a supervised learning task where the objective is to categorize instances (examples) into one of **two discrete classes**.
- A **multiclass classification** task is a supervised learning problem where the objective is to categorize instances into one of **three or more discrete classes**.

Caveat: Multiclass classification should not be confused with **multilabel classification**, which allows an instance to be associated with multiple classes.

- An example of a binary classification task is the prediction of disease status using genomic data (Wu et al. 2018).
- Researchers often seek to accurately classify the type, subtype, or stage of cancer using gene expression data. For example, in the Pan-Cancer Atlas, specific cancers such as Breast Cancer (BRCA), Colon Adenocarcinoma (COAD), Lung Adenocarcinoma (LUAD), Ovarian Cancer (OV), and Thyroid Cancer (THCA) can be

differentiated based on their unique gene expression profiles (Alharbi and Vakanski 2023).

Binary classification

- Some machine learning algorithms are specifically designed to solve binary classification problems.
 - **Logistic regression** and **support vector machines** (SVMs) are such examples.

Later in the presentation, make sure to understand why the logistic regression is specifically designed to solve binary classification problems.

Multiclass classification

- A **multiclass classification problem** can be approached as a collection of **binary classification tasks**.
- **One-vs-Rest (OvR)**
 - A **separate binary classifier** is trained **for each class**.
 - For each classifier, **one class** is treated as the **positive class**, and **all other classes** are treated as the **negative class**.
 - The **final assignment** selects the class whose classifier produces the **largest score** for a given input.

Scikit-learn offers `OneVsRestClassifier`, a utility designed to extend binary classifiers to multiclass tasks. One-vs-rest is also called one-vs-all.

A complete example will be presented at the end of the lecture.

Logistic Regression

Data and Problem

- **Dataset:** Palmer Penguins
- **Task:** Binary classification to distinguish Gentoo penguins from non-Gentoo species
- **Feature of Interest:** Flipper length

Histogram

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from palmerpenguins import load_penguins
```

```

from sklearn.datasets import load_digits, load_iris
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from sklearn.multiclass import OneVsRestClassifier
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler, label_binarize

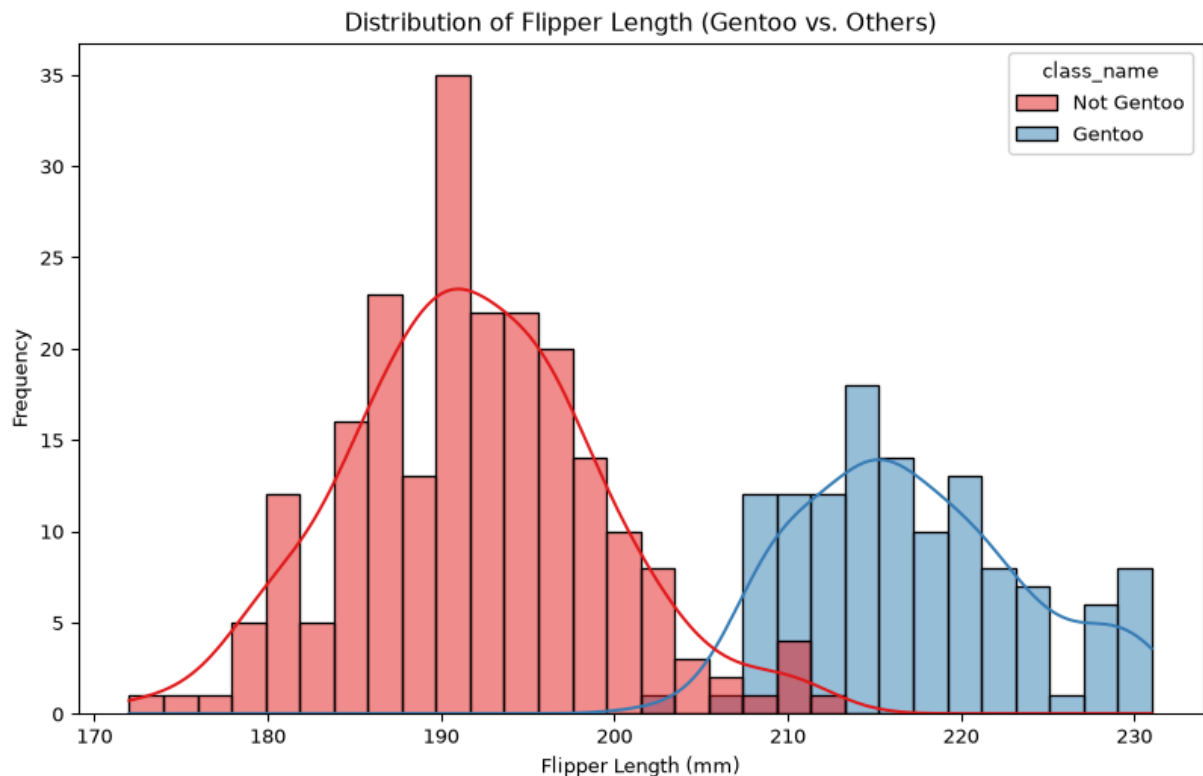
# Load the Palmer Penguins dataset
penguin_df = load_penguins()[['flipper_length_mm', 'species']].dropna().copy

# Create a binary label: 1 if Gentoo, 0 otherwise
penguin_df['is_gentoo'] = (penguin_df['species'] == 'Gentoo').astype(int)
penguin_df['class_name'] = np.where(
    penguin_df['is_gentoo'] == 1,
    'Gentoo',
    'Not Gentoo'
)

# Separate features (X) and labels (y)
penguin_X = penguin_df[['flipper_length_mm']]
penguin_y = penguin_df['is_gentoo']

# Plot the distribution of flipper lengths by binary species label
plt.figure(figsize=(10, 6))
sns.histplot(
    data=penguin_df,
    x='flipper_length_mm',
    hue='class_name',
    hue_order=['Not Gentoo', 'Gentoo'],
    kde=True,
    bins=30,
    palette='Set1'
)
plt.title('Distribution of Flipper Length (Gentoo vs. Others)')
plt.xlabel('Flipper Length (mm)')
plt.ylabel('Frequency')
plt.show()

```



Logistic (Logit) Regression

- Despite its name, **logistic regression** serves as a **classification** algorithm rather than a regression technique.
- The **labels** in logistic regression are binary values, denoted as $y_i \in \{0, 1\}$, making it a **binary classification task**.
- The primary objective of logistic regression is to determine the **probability** that a given instance x_i belongs to the **positive class**, i.e., $y_i = 1$.

The representation of the two classes, **negative** and **positive**, by the values 0 and 1, respectively, is not arbitrary. This choice is intrinsically connected to our objective of determining the **probability** that an instance x_i belongs to the positive class.

While this learning algorithm may initially seem unremarkable, it is essential to continue engaging with it, as logistic regression will later prove to be crucial in the discussion on artificial neural networks.

Model

- **General Case:** $P(y = k|x, \theta)$, where k is a class label.
- **Binary Case:** $y \in \{0, 1\}$
 - **Predict** $P(y = 1|x, \theta)$

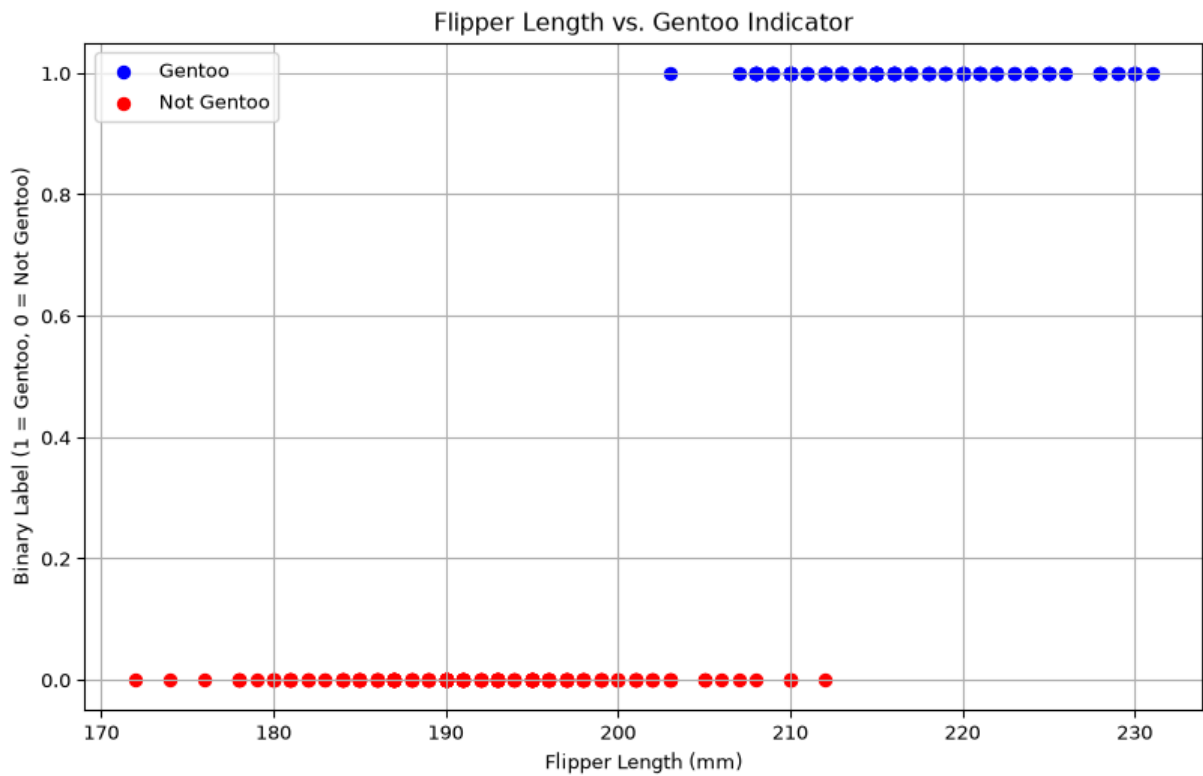
Visualizing our data

```
In [3]: # Scatter plot of flipper length vs. binary label (Gentoo or Not Gentoo)
plt.figure(figsize=(10, 6))

# Plot points labeled as Gentoo (is_gentoo = 1)
plt.scatter(
    penguin_df.loc[penguin_df['is_gentoo'] == 1, 'flipper_length_mm'],
    penguin_df.loc[penguin_df['is_gentoo'] == 1, 'is_gentoo'],
    color='blue',
    label='Gentoo'
)

# Plot points labeled as Not Gentoo (is_gentoo = 0)
plt.scatter(
    penguin_df.loc[penguin_df['is_gentoo'] == 0, 'flipper_length_mm'],
    penguin_df.loc[penguin_df['is_gentoo'] == 0, 'is_gentoo'],
    color='red',
    label='Not Gentoo'
)

plt.title('Flipper Length vs. Gentoo Indicator')
plt.xlabel('Flipper Length (mm)')
plt.ylabel('Binary Label (1 = Gentoo, 0 = Not Gentoo)')
plt.legend(loc='best')
plt.grid(True)
plt.show()
```



Intuition

Fitting a linear regression is not the answer, but ...

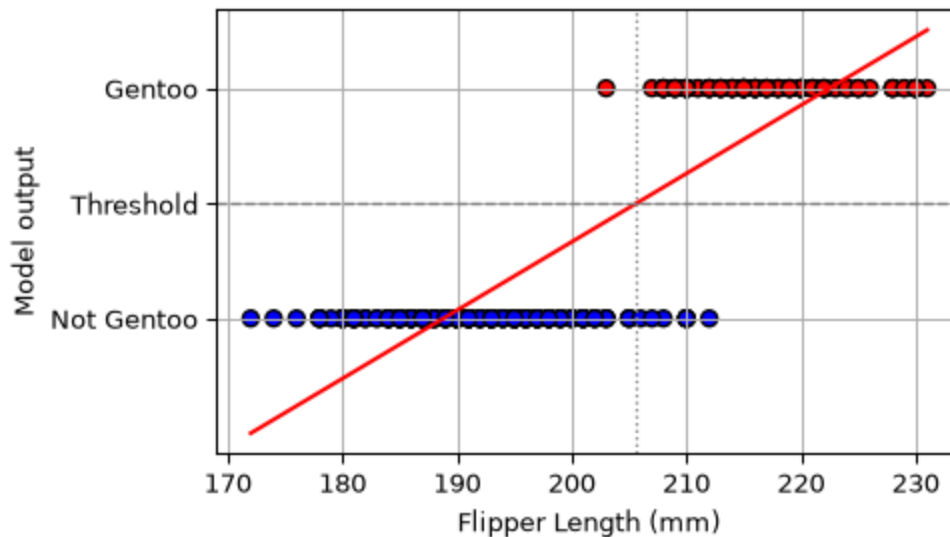
```
In [4]: penguin_linear_model = LinearRegression()
penguin_linear_model.fit(penguin_X, penguin_y)

penguin_line_X = pd.DataFrame({
    'flipper_length_mm': np.linspace(
        penguin_X['flipper_length_mm'].min(),
        penguin_X['flipper_length_mm'].max(),
        200
    )
})
penguin_line_y = penguin_linear_model.predict(penguin_line_X)

# The threshold is where the fitted line reaches 0.5.
linear_boundary_mm = float(
    (0.5 - penguin_linear_model.intercept_) /
    penguin_linear_model.coef_[0]
)

def plot_penguin_linear_fit():
    plt.figure(figsize=(5, 3))
    plt.scatter(
        penguin_X['flipper_length_mm'],
        penguin_y,
        c=penguin_y,
        cmap='bwr',
        edgecolor='k'
    )
    plt.plot(penguin_line_X['flipper_length_mm'], penguin_line_y, 'r-')
    plt.axhline(0.5, color='gray', linestyle='--', linewidth=1)
    plt.axvline(linear_boundary_mm, color='gray', linestyle=':', linewidth=1)
    plt.xlabel('Flipper Length (mm)')
    plt.ylabel('Model output')
    plt.yticks([0, 0.5, 1], ['Not Gentoo', 'Threshold', 'Gentoo'])
    plt.grid(True)
    plt.show()

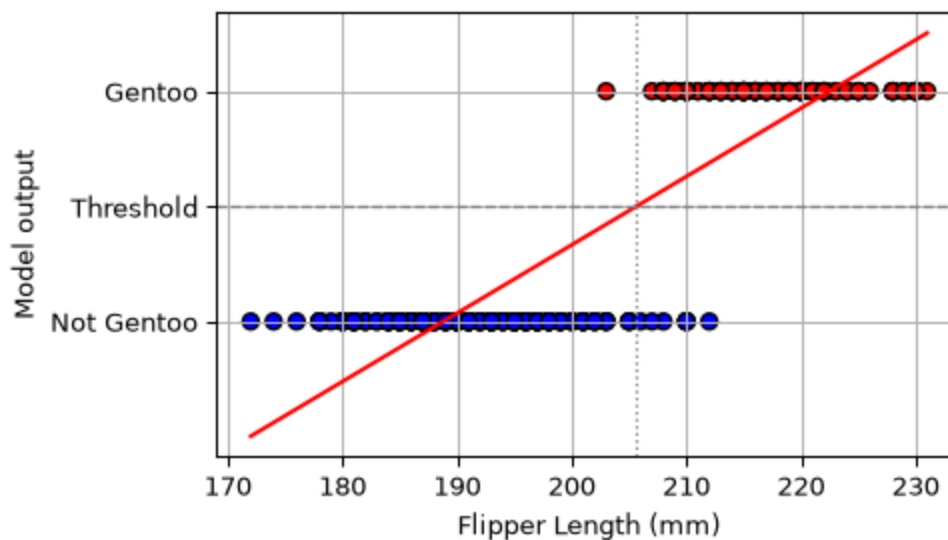
plot_penguin_linear_fit()
```



```
In [5]: plot_penguin_linear_fit()
```

Intuition (continued)

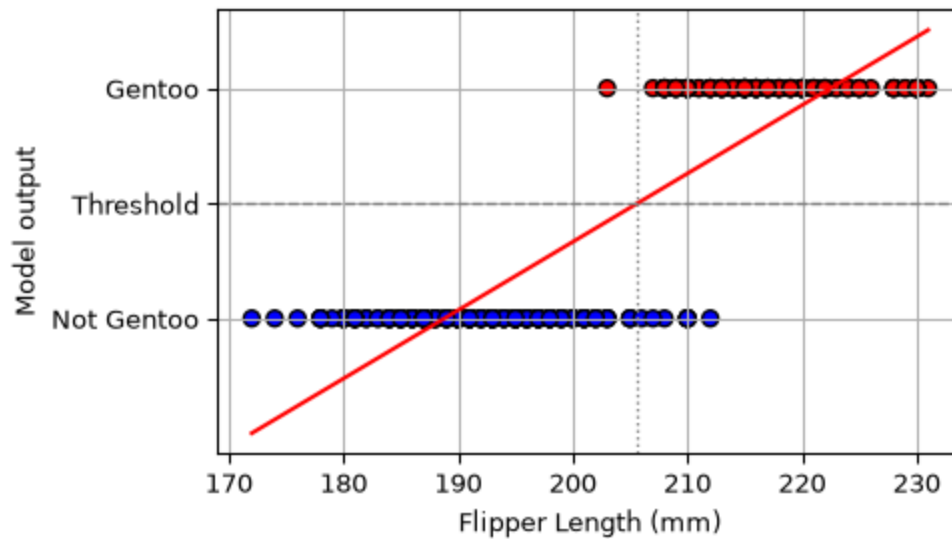
```
plot_penguin_linear_fit()
```



- A high `flipper_length_mm` typically results in a model output approaching 1.
- Conversely, a low `flipper_length_mm` generally yields a model output near 0.
- Notably, the model outputs are not confined to the $[0, 1]$ interval and may occasionally fall below 0 or surpass 1.

Intuition (continued)

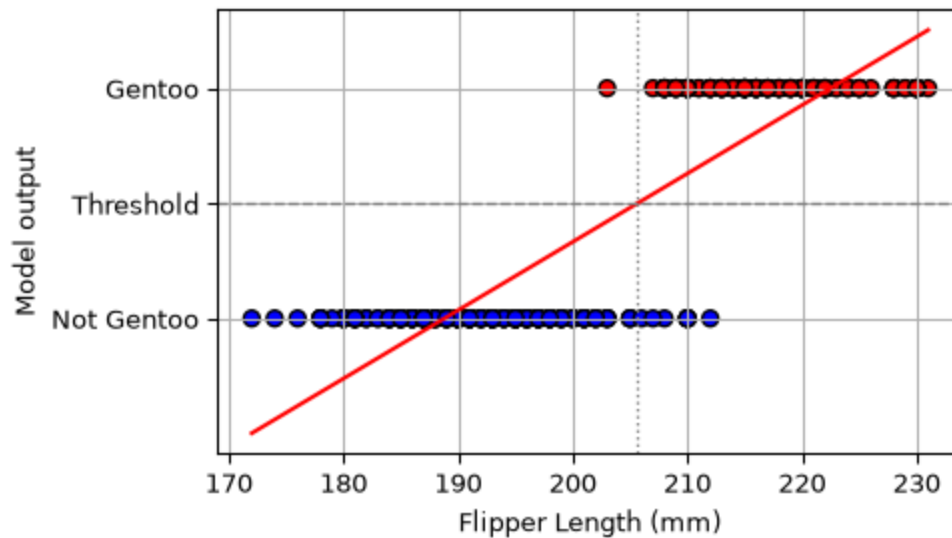
```
plot_penguin_linear_fit()
```



- For a **single feature**, the decision boundary is a **specific point**.
- In this case, the fitted line reaches the 0.5 threshold at approximately 205.6 mm.

Intuition (continued)

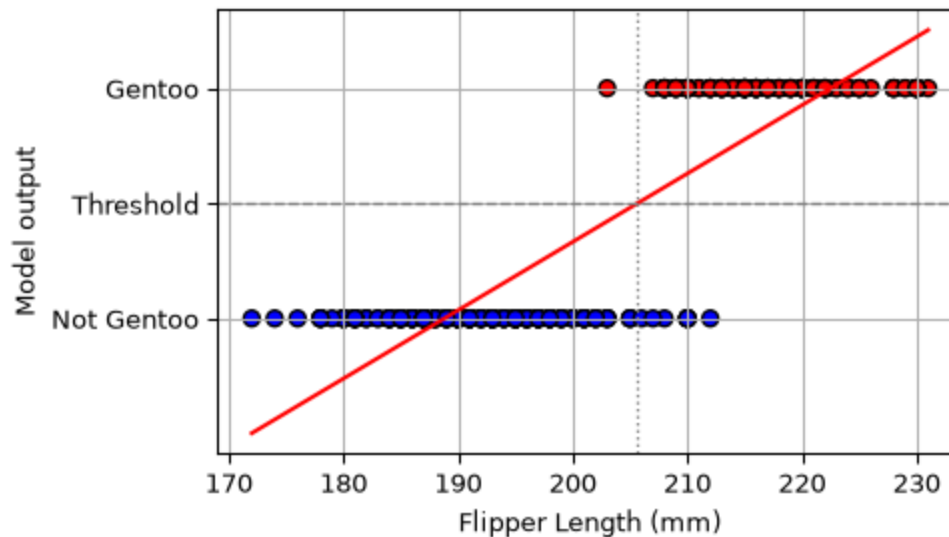
`plot_penguin_linear_fit()`



- As `flipper_length_mm` increases above the threshold, the fitted output moves toward the Gentoo label.
- As `flipper_length_mm` decreases below the threshold, the fitted output moves toward the non-Gentoo label.

Intuition (continued)

`plot_penguin_linear_fit()`



- Near the threshold, the fitted output is close to 0.5, but linear regression still does not produce a valid probability model.

Logistic Function

```
In [10]: def sigmoid(t):
    return 1 / (1 + np.exp(-t))

sigmoid_input = np.linspace(-6, 6, 1000)
sigmoid_output = sigmoid(sigmoid_input)

# Fit a one-feature logistic regression model so that we can distinguish
# its linear score from the probability obtained after applying the sigmoid.
penguin_logistic_model = LogisticRegression(max_iter=1000)
penguin_logistic_model.fit(penguin_X, penguin_y)

penguin_line_score = penguin_logistic_model.decision_function(penguin_line_X)
logistic_boundary_mm = float(
    -penguin_logistic_model.intercept_[0] /
    penguin_logistic_model.coef_[0, 0]
)

def plot_penguin_logistic_score():
    plt.figure(figsize=(5, 3))
    plt.plot(
        penguin_line_X['flipper_length_mm'],
        penguin_line_score,
        color='blue',
        linewidth=2
    )
    plt.axhline(0, color='gray', linestyle='--', linewidth=1)
    plt.axvline(logistic_boundary_mm, color='gray', linestyle=':', linewidth=1)
    plt.scatter([logistic_boundary_mm], [0], color='black', zorder=3)
    plt.xlabel('Flipper Length (mm)')
    plt.ylabel(r'Linear score  $t(x)$ ')
```

```
plt.grid(True)
plt.show()

def plot_sigmoid():
    fig, ax = plt.subplots(figsize=(6, 4))
    ax.plot(sigmoid_input, sigmoid_output, color='blue', linewidth=2)
    ax.axvline(x=0, color='black', linewidth=1)
    ax.axhline(y=0.5, color='gray', linestyle='--', linewidth=1)
    ax.scatter([0], [0.5], color='black', zorder=3)
    ax.set_yticks([0, 0.5, 1.0])
    ax.set_xlabel('t')
    ax.set_ylabel(r'\sigma(t)')
    ax.grid(True)
    plt.show()

plot_sigmoid()
```

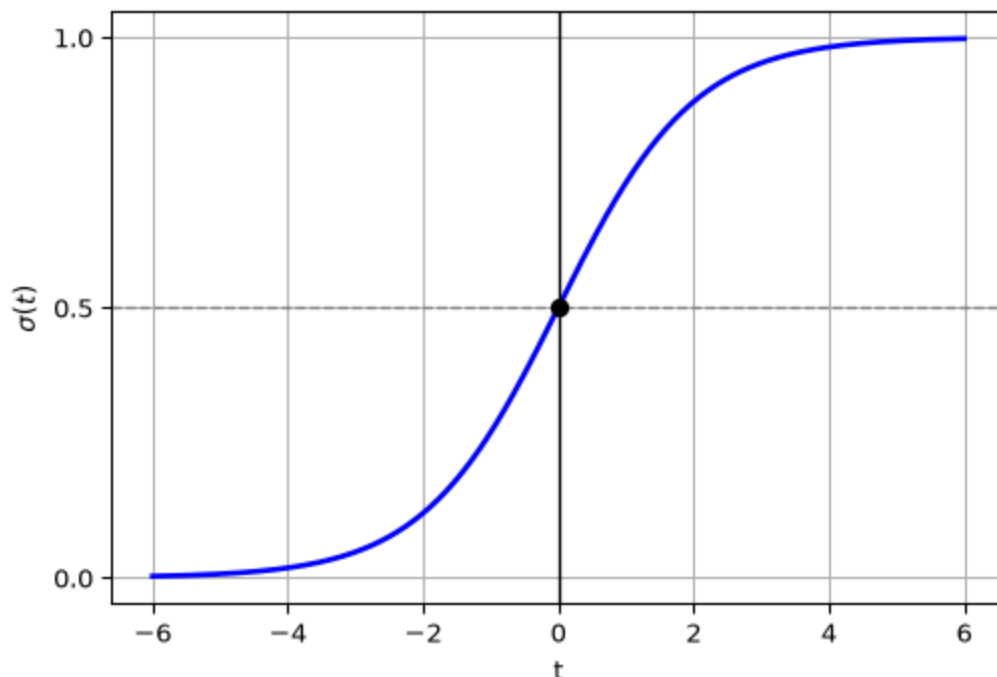
In [11]: plot_sigmoid()

Logistic Function

In mathematics, the **standard logistic function** maps a real-valued input from $\mathbb{R}$ to the open interval $(0, 1)$. The function is defined as:

$$\sigma(t) = \frac{1}{1 + e^{-t}}$$

plot_sigmoid()

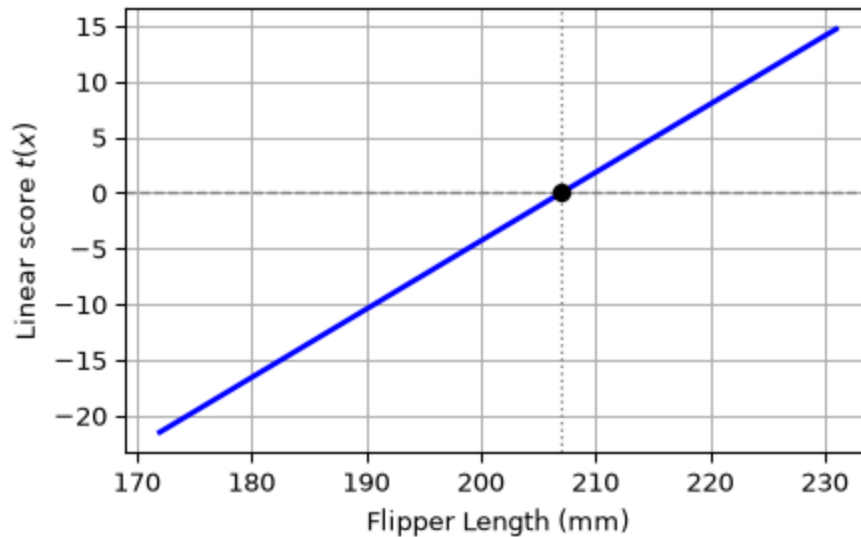


- When the input variable t is 0, the output of the logistic function is 0.5.
 - Indeed, $e^{-t} = e^0 = 1$ and thus $\frac{1}{1+e^{-t}} = \frac{1}{1+1} = \frac{1}{2}$.

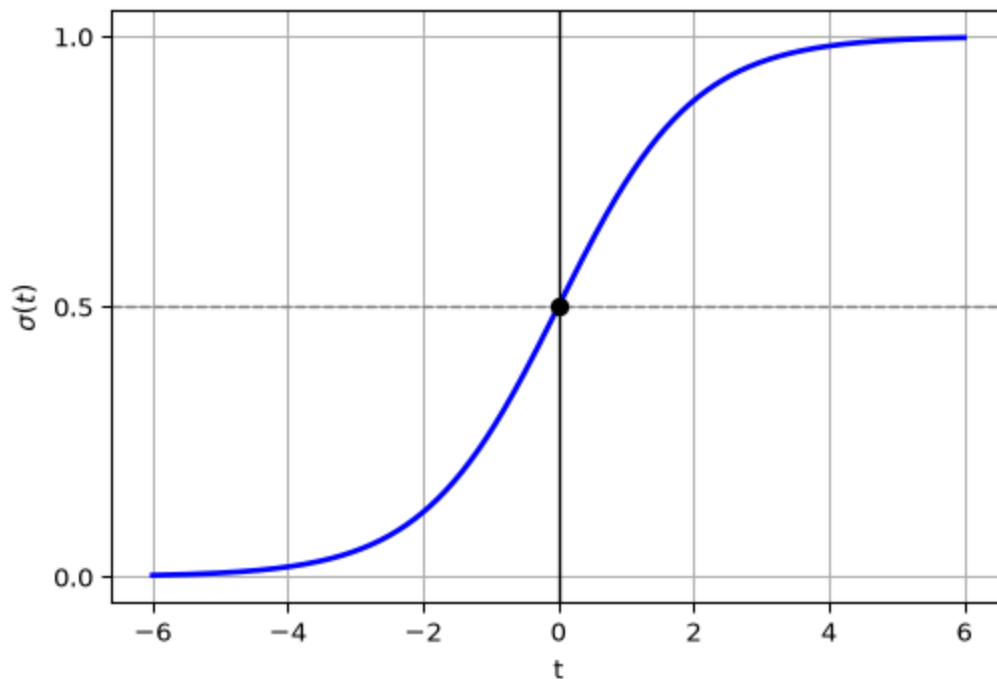
- As t increases, the output value approaches 1.
 - As $t \rightarrow \infty, e^{-t} \rightarrow 0$.
- Conversely, as t becomes more negative, the output value approaches 0.
 - As $t \rightarrow -\infty, e^{-t} \rightarrow \infty$ (the two negative signs cancel out).
- The standard logistic function is also commonly referred to as the **sigmoid** function.

Linear score and probability

`plot_penguin_logistic_score()`



`plot_sigmoid()`



- The fitted linear score $t(x) = \theta_0 + \theta_1 x$ is zero at the decision boundary, approximately 207 mm in this example.

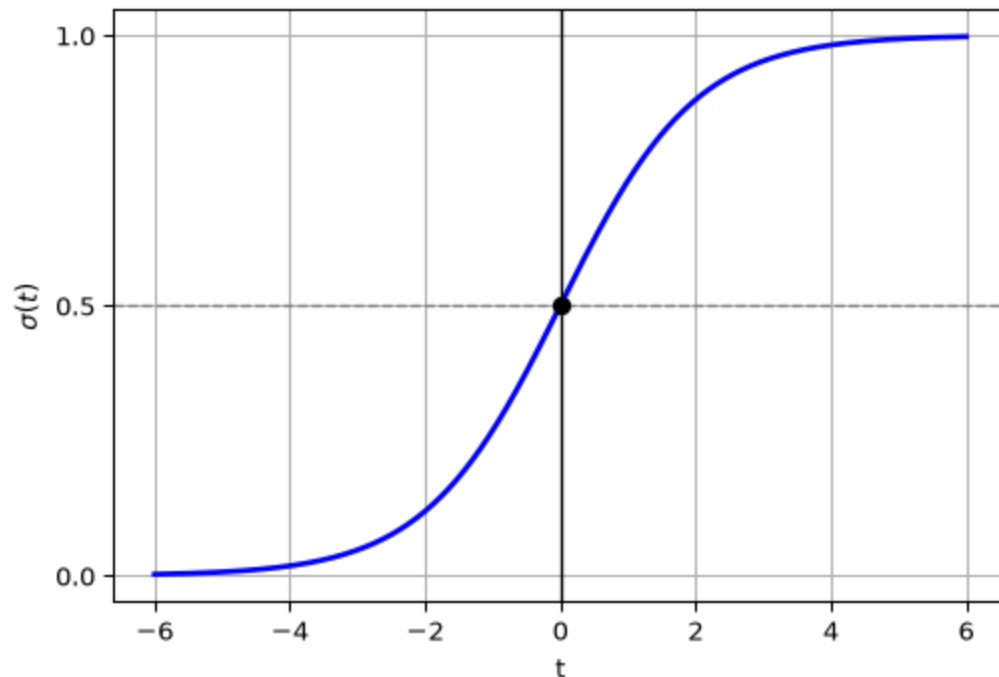
- The sigmoid maps this zero score to $P(y = 1 \mid x) = 0.5$. Negative scores map below 0.5, and positive scores map above 0.5.

Logistic function

An S-shaped curve, such as the standard logistic function (aka **sigmoid**), is termed a **squashing function** because it maps a wide input domain to a constrained output range.

$$\sigma(t) = \frac{1}{1 + e^{-t}}$$

plot_sigmoid()



Logistic (Logit) Regression

- Analogous to **linear regression**, **logistic regression** computes a weighted sum of the input features, expressed as:

$$\theta_0 + \theta_1 x_i^{(1)} + \theta_2 x_i^{(2)} + \dots + \theta_D x_i^{(D)}$$

- However, using the sigmoid function limits its output to the range $(0, 1)$:

$$\sigma(\theta_0 + \theta_1 x_i^{(1)} + \theta_2 x_i^{(2)} + \dots + \theta_D x_i^{(D)})$$

Notation

- Equation for the **logistic regression**:

$$\sigma(\theta_0 + \theta_1 x_i^{(1)} + \theta_2 x_i^{(2)} + \dots + \theta_D x_i^{(D)})$$

- Multiplying θ_0 (intercept/bias) by 1:

$$\sigma(\theta_0 \times 1 + \theta_1 x_i^{(1)} + \theta_2 x_i^{(2)} + \dots + \theta_D x_i^{(D)})$$

- Multiplying θ_0 by $x_i^{(0)} = 1$:

$$\sigma(\theta_0 x_i^{(0)} + \theta_1 x_i^{(1)} + \theta_2 x_i^{(2)} + \dots + \theta_D x_i^{(D)})$$

Logistic regression

The **Logistic Regression** model, in its vectorized form, is defined as:

$$h_{\theta}(x_i) = \sigma(\theta^{\top} x_i) = \frac{1}{1 + e^{-\theta^{\top} x_i}}$$

An extra feature, $x_i^{(0)} = 1$, has been added to x_i , and θ_0 is the intercept/bias term.

```
In [16]: penguin_features = ['bill_depth_mm', 'body_mass_g']
penguin_two_feature_df = load_penguins()[penguin_features + ['species']].drop
penguin_two_feature_df['is_gentoo'] = (
    penguin_two_feature_df['species'] == 'Gentoo'
).astype(int)

penguin_two_X = penguin_two_feature_df[penguin_features].to_numpy()
penguin_two_y = penguin_two_feature_df['is_gentoo'].to_numpy()

penguin_two_X_train, penguin_two_X_test, penguin_two_y_train, penguin_two_y_
    penguin_two_X,
    penguin_two_y,
    test_size=0.2,
    random_state=42,
    stratify=penguin_two_y
)

penguin_two_model = make_pipeline(
    StandardScaler(),
    LogisticRegression(max_iter=1000)
)
penguin_two_model.fit(penguin_two_X_train, penguin_two_y_train)

def plot_penguin_decision_boundary(X_values, y_values, fitted_model):
    bill_depth = np.linspace(X_values[:, 0].min() - 1, X_values[:, 0].max()
    body_mass = np.linspace(X_values[:, 1].min() - 200, X_values[:, 1].max()
    depth_grid, mass_grid = np.meshgrid(bill_depth, body_mass)
    grid = np.column_stack([depth_grid.ravel(), mass_grid.ravel()])
    predicted_class = fitted_model.predict(grid).reshape(depth_grid.shape)

    plt.figure(figsize=(9, 4))
```

```

plt.contourf(depth_grid, mass_grid, predicted_class, alpha=0.3, cmap='Rd
plt.scatter(
    X_values[y_values == 1, 0],
    X_values[y_values == 1, 1],
    color='orange',
    edgecolors='k',
    label='Gentoo'
)
plt.scatter(
    X_values[y_values == 0, 0],
    X_values[y_values == 0, 1],
    color='blue',
    edgecolors='k',
    label='Not Gentoo'
)
plt.xlabel('Bill Depth (mm)')
plt.ylabel('Body Mass (g)')
plt.title('Logistic Regression Decision Regions')
plt.legend()
plt.show()

plot_penguin_decision_boundary(
    penguin_two_X_train,
    penguin_two_y_train,
    penguin_two_model
)

```

```

In [17]: plot_penguin_decision_boundary(
    penguin_two_X_train,
    penguin_two_y_train,
    penguin_two_model
)

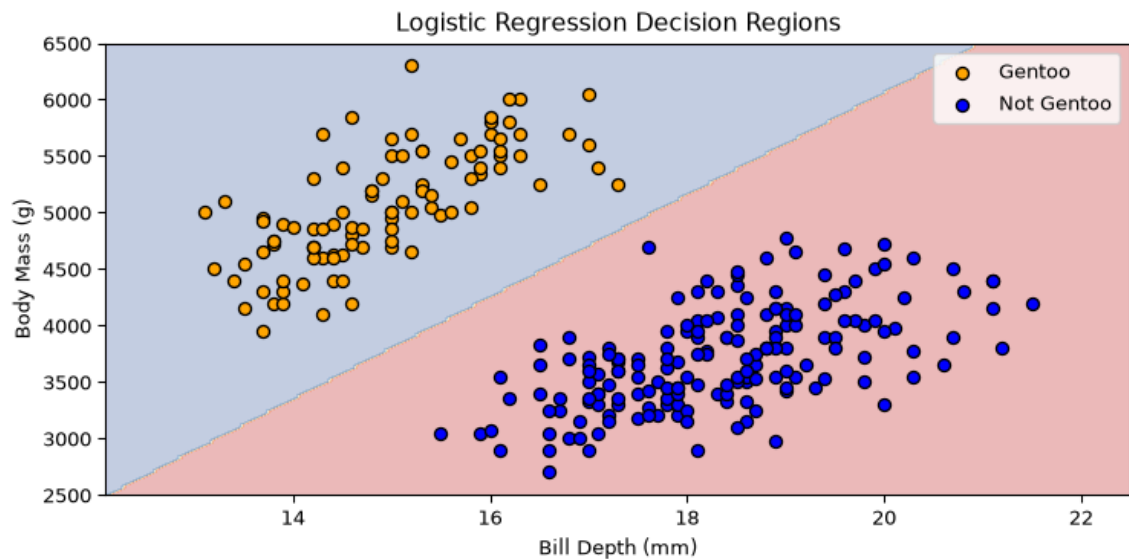
```

Logistic regression (two attributes)

```

plot_penguin_decision_boundary(
    penguin_two_X_train,
    penguin_two_y_train,
    penguin_two_model
)

```



$$h_{\theta}(x_i) = \sigma(\theta^{\top} x_i)$$

- The sign of $\theta^{\top} x_i$ determines which side of the decision boundary contains the example.
- As the score becomes more positive or more negative, the model probability moves farther from 0.5.
- On the decision boundary, $\theta^{\top} x_i = 0$ and the model assigns probability 0.5 to each class.

Logistic regression

- The **Logistic Regression** model, in its vectorized form, is defined as:

$$h_{\theta}(x_i) = \sigma(\theta^{\top} x_i) = \frac{1}{1 + e^{-\theta^{\top} x_i}}$$

- **Predictions** are made as follows:

- $\hat{y}_i = 0$, if $h_{\theta}(x_i) < 0.5$, equivalently $\theta^{\top} x_i < 0$
- $\hat{y}_i = 1$, if $h_{\theta}(x_i) \geq 0.5$, equivalently $\theta^{\top} x_i \geq 0$

...

- The parameters θ are learned by minimizing a loss function. The next lecture derives this loss and applies gradient descent.

<https://youtu.be/yIYKR4sgzI8>

Attribution: StatQuest: Logistic Regression by Josh Starmer.

"StatQuest breaks down complicated Statistics and Machine Learning methods into small, bite-sized pieces that are easy to understand. StatQuest doesn't dumb down the material, instead, it builds you up so that you are smarter and have a better understanding of Statistics and Machine Learning." It is often a good place to start to get the intuition to understand statistical and machine learning concepts.

Digits example

1989 Yann LeCun

<https://www.youtube.com/watch?v=H0oEr40YhrQ>

Handwritten Digit Recognition

Aims:

- Developing a **logistic regression model** for the **recognition of handwritten digits**.
- Visualize the **insights** and **patterns** the model has acquired.

We are using the [UCI ML hand-written digits datasets](#).

UCI ML hand-written digits datasets

Loading the dataset

```
In [19]: digits = load_digits()
```

What is the type of `digits.data`

```
In [20]: type(digits.data)
```

`numpy.ndarray`

Developing a **logistic regression model** for the **recognition of handwritten digits**.

UCI ML hand-written digits datasets

How many examples (`N`) and how many attributes (`D`)?

```
In [21]: digits.data.shape
```

`(1797, 64)`

Assigning `N` and `D`

```
In [22]: digit_count, digit_feature_count = digits.data.shape
```

`target` has the same number of entries (examples) as `data` ?

```
In [23]: digits.target.shape
```

(1797,)

UCI ML hand-written digits datasets

What are the width and height of those images?

```
In [24]: digits.images.shape
```

(1797, 8, 8)

Assigning `height` and `width`

```
In [25]: _, digit_height, digit_width = digits.images.shape
```

UCI ML hand-written digits datasets

Assigning `digits_X` and `digits_y`

```
In [26]: digits_X = digits.data
         digits_y = digits.target
```

UCI ML hand-written digits datasets

`digits_X[0]` is a vector of size `digit_height * digit_width = D` ($8 \times 8 = 64$).

```
In [27]: digits_X[0]
```

```
array([ 0.,  0.,  5., 13.,  9.,  1.,  0.,  0.,  0.,  0., 13., 15., 10.,
        15.,  5.,  0.,  0.,  3., 15.,  2.,  0., 11.,  8.,  0.,  0.,  4.,
        12.,  0.,  0.,  8.,  8.,  0.,  0.,  5.,  8.,  0.,  0.,  9.,  8.,
         0.,  0.,  4., 11.,  0.,  1., 12.,  7.,  0.,  0.,  2., 14.,  5.,
        10., 12.,  0.,  0.,  0.,  0.,  6., 13., 10.,  0.,  0.,  0.]
```

...

It corresponds to an $8 \times 8 = 64$ image.

```
In [28]: digits_X[0].reshape(digit_height, digit_width)
```

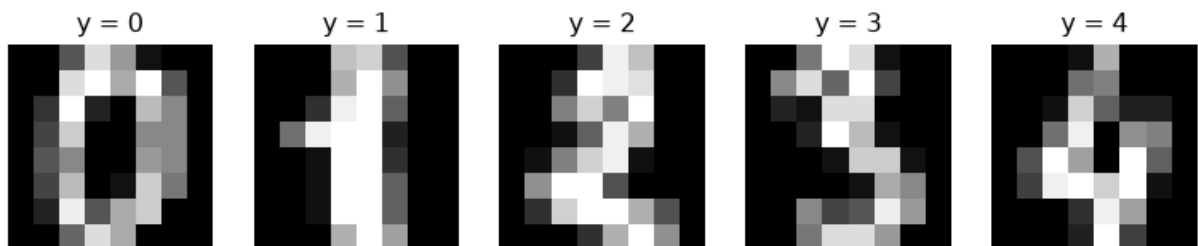
```
array([[ 0.,  0.,  5., 13.,  9.,  1.,  0.,  0.],
       [ 0.,  0., 13., 15., 10., 15.,  5.,  0.],
       [ 0.,  3., 15.,  2.,  0., 11.,  8.,  0.],
       [ 0.,  4., 12.,  0.,  0.,  8.,  8.,  0.],
       [ 0.,  5.,  8.,  0.,  0.,  9.,  8.,  0.],
       [ 0.,  4., 11.,  0.,  1., 12.,  7.,  0.],
       [ 0.,  2., 14.,  5., 10., 12.,  0.,  0.],
       [ 0.,  0.,  6., 13., 10.,  0.,  0.,  0.]])
```

UCI ML hand-written digits datasets

Plot the first `n=5` examples

```
In [29]: def plot_digit_examples(X_values, y_values, count=5):
plt.figure(figsize=(10, 2))
for index, (image, label) in enumerate(zip(X_values[:count], y_values[:count])):
    plt.subplot(1, count, index + 1)
    plt.imshow(image.reshape(digit_height, digit_width), cmap=plt.cm.gray)
    plt.title(f'y = {label}')
    plt.axis('off')
plt.show()

plot_digit_examples(digits_X, digits_y)
```

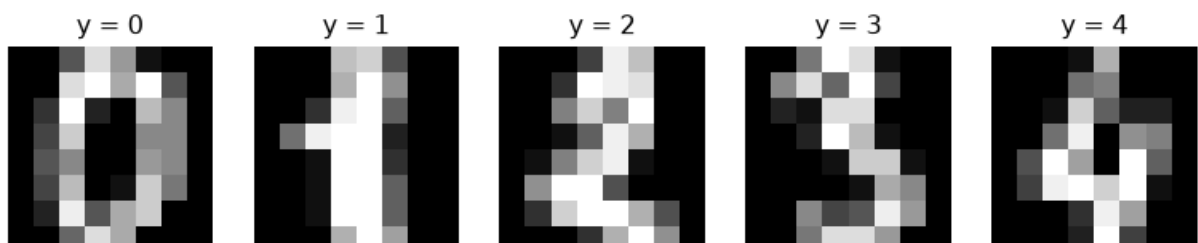


Intensity values close to 0 are represented in black, while high values are represented in white.

Use `cmap=plt.cm.binary` to invert the images (intensity values close to 0 in white, high values in black).

UCI ML hand-written digits datasets

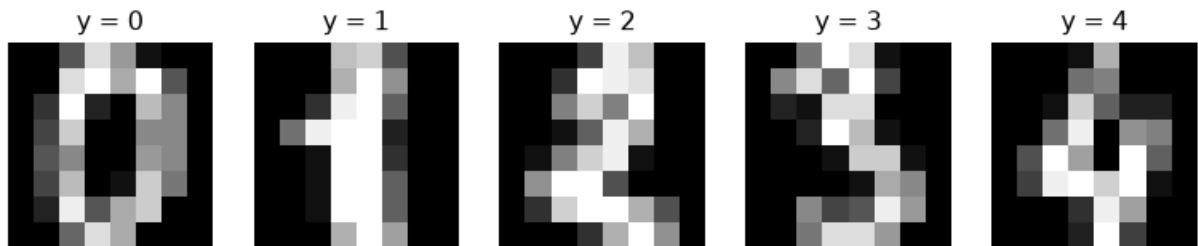
```
In [30]: plot_digit_examples(digits_X, digits_y)
```



- In our dataset, each x_i is an **attribute vector** of size $D = 64$.
- This vector is formed by concatenating the rows of an 8×8 image.
- The `reshape` function is employed to convert this 64-dimensional vector back into its original 8×8 image format.

UCI ML hand-written digits datasets

In [31]: `plot_digit_examples(digits_X, digits_y)`



- We will train **10 classifiers**, each corresponding to a specific digit in a **one-vs-rest (OvR)** approach.
- Each classifier will determine the optimal values of θ_j (associated with the pixel features), allowing it to distinguish one digit from all other digits.

UCI ML hand-written digits datasets

Preparing for our machine learning experiment

```
In [32]: digits_X_train, digits_X_test, digits_y_train, digits_y_test = train_test_split(
    digits_X,
    digits_y,
    test_size=0.1,
    random_state=42,
    stratify=digits_y
)
```

UCI ML hand-written digits datasets

Optimization algorithms generally work best when the attributes have similar ranges.

```
In [33]: digit_scaler = StandardScaler()
    digits_X_train_scaled = digit_scaler.fit_transform(digits_X_train)
    digits_X_test_scaled = digit_scaler.transform(digits_X_test)
```

Discussion: importance of applying `fit_transform` only to `digits_X_train`.

Optimization generally behaves better when the features have comparable scales.

[StandardScaler](#) is a commonly used approach. For a given feature, the values are transformed as follows:

$$z = (x - \mu) / \sigma$$

where μ is the average of all the values for that feature for the training data, and σ the corresponding standard deviation.

Calling `fit` learns the mean and standard deviation of each feature. Calling `transform` applies the learned transformation. Calling `fit_transform` combines those operations.

Fitting the scaler on the complete dataset would cause **data leakage** because information from the test set would influence the transformation used during training.

The test data must still be transformed, but with the mean and standard deviation learned from the training data. The fitted scaler is therefore part of the trained system.

UCI ML hand-written digits datasets

```
In [34]: digit_classifier = OneVsRestClassifier(LogisticRegression(max_iter=1000))
digit_classifier.fit(digits_X_train_scaled, digits_y_train)
```

The classifier solves the multiclass task using **one-vs-rest (OvR)**. Classifiers in scikit-learn provide [multiclass support](#).

Each logistic regression finds a **hyperplane** in a **64-dimensional space** that separates the data into two classes.

UCI ML hand-written digits datasets

Applying the classifier to our test set

```
In [35]: digits_y_pred = digit_classifier.predict(digits_X_test_scaled)
digit_test_accuracy = accuracy_score(digits_y_test, digits_y_pred)
print(f'Test accuracy: {digit_test_accuracy:.3f}')
```

Test accuracy: 0.967

Lecture 7 introduces performance measures in detail.

Visualization

How many classes?

```
In [36]: digit_classifier.classes_
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
...
```

The coefficients and intercepts are in distinct arrays.

```
In [37]: (digit_classifier.estimators_[0].coef_.shape,
          digit_classifier.estimators_[0].intercept_.shape)
((1, 64), (1,))
...
```

The intercept is θ_0 . The 64 feature coefficients are θ_j , for $j \in \{1, \dots, 64\}$.

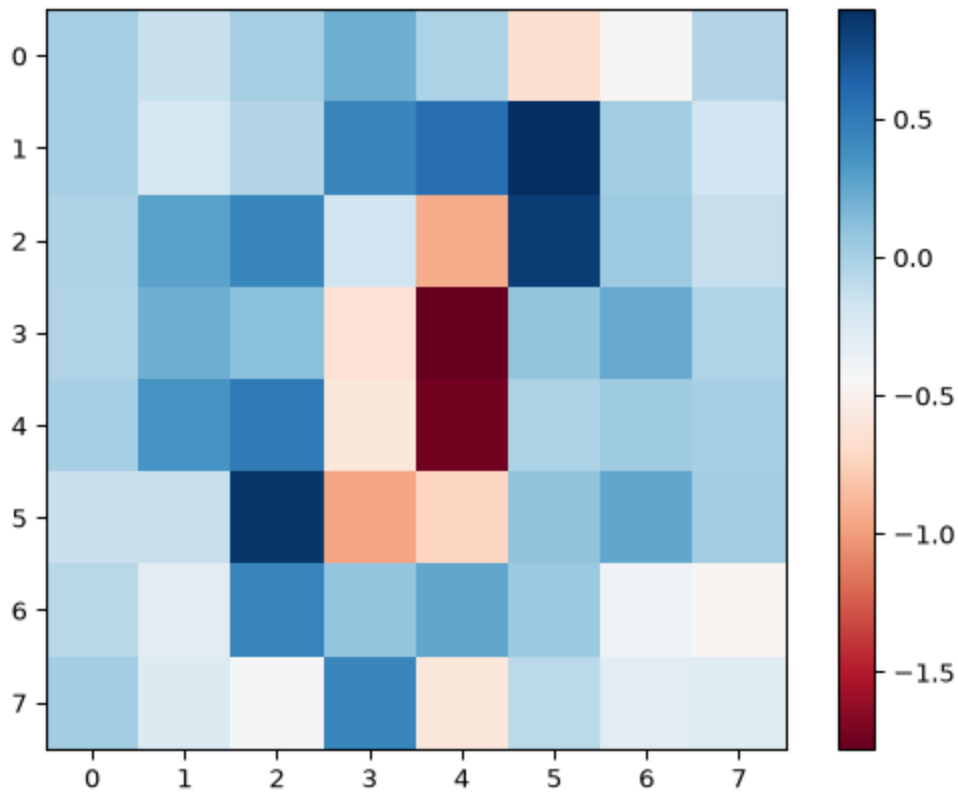
Adapted from [MNIST digits classification using Logistic regression in Scikit-Learn](#),
visited 2024-09-23.

Visualization

```
In [38]: digit_classifier.estimators_[0].coef_[0].round(2).reshape(
          digit_height,
          digit_width
        )
array([[ 0.   , -0.14, -0.   ,  0.21, -0.01, -0.66, -0.46, -0.05],
       [ 0.   , -0.22, -0.05,  0.43,  0.57,  0.9 ,  0.02, -0.19],
       [-0.03,  0.28,  0.43, -0.18, -0.94,  0.82,  0.04, -0.13],
       [-0.04,  0.21,  0.11, -0.63, -1.78,  0.09,  0.23, -0.04],
       [ 0.   ,  0.35,  0.5 , -0.6 , -1.73, -0.02,  0.03,  0.   ],
       [-0.15, -0.13,  0.86, -0.97, -0.73,  0.1 ,  0.26,  0.02],
       [-0.07, -0.3 ,  0.44,  0.09,  0.25,  0.05, -0.4 , -0.47],
       [ 0.02, -0.26, -0.43,  0.43, -0.59, -0.08, -0.28, -0.28]])
```

Visualization

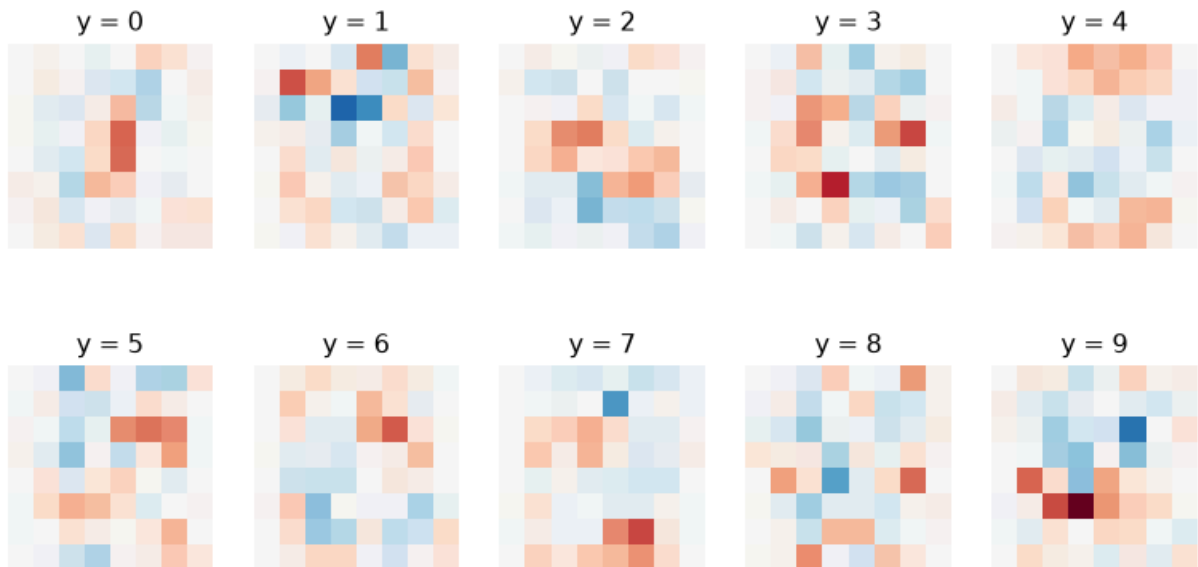
```
In [39]: digit_zero_coefficients = digit_classifier.estimators_[0].coef_[0]
plt.imshow(
    digit_zero_coefficients.reshape(digit_height, digit_width),
    cmap=plt.cm.RdBu
)
plt.colorbar()
plt.show()
```



Visualization

```
In [40]: plt.figure(figsize=(10,5))
max_abs_coefficient = max(
    np.abs(estimator.coef_).max()
    for estimator in digit_classifier.estimators_
)

for index, class_label in enumerate(digit_classifier.classes_):
    plt.subplot(2, 5, index + 1)
    plt.title(f'y = {class_label}')
    plt.imshow(
        digit_classifier.estimators_[index].coef_.reshape(
            digit_height,
            digit_width
        ),
        cmap=plt.cm.RdBu,
        vmin=-max_abs_coefficient,
        vmax=max_abs_coefficient
    )
    plt.axis('off')
plt.show()
```



Each binary `LogisticRegression` learns 64 feature coefficients and one intercept, for 65 parameters.

These coefficients act on the standardized pixel intensities used for training.

Red pixels correspond to negative coefficients, while blue pixels correspond to positive coefficients. All ten maps use the same colour scale.

For the classifier that recognizes zero, high-intensity pixels in the centre reduce the class score, while pixels along the surrounding oval increase it.

<https://youtu.be/AX-ZEC-71DI>

Attribution: [Machine Learning and Logistic Regression](#), IBM Technology, 2024-07-19.

Prologue

Summary

- Logistic regression maps the linear score $\theta^\top x_i$ to the probability $P(y_i = 1 \mid x_i, \theta)$ using the sigmoid function.
- The threshold $h_\theta(x_i) = 0.5$ corresponds to the linear decision boundary $\theta^\top x_i = 0$.
- A model probability estimates class membership; it is not the probability that the predicted label is correct.
- **One-vs-rest** trains one binary classifier for each class and selects the class with the largest score.
- Preprocessing must be fitted on the training set and then applied unchanged to the test set.
- Learned coefficients show how each feature changes a classifier's linear score.

References

Alharbi, Fadi, and Aleksandar Vakanski. 2023. "Machine Learning Methods for Cancer Classification Using Gene Expression Data: A Review." *Bioengineering* 10 (2): 173. <https://doi.org/10.3390/bioengineering10020173>.

Russell, Stuart, and Peter Norvig. 2020. *Artificial Intelligence: A Modern Approach*. 4th ed. Pearson. <http://aima.cs.berkeley.edu/>.

Wu, Qianfan, Adel Boueiz, Alican Bozkurt, et al. 2018. "Deep Learning Methods for Predicting Disease Status Using Genomic Data." *Journal of Biometrics & Biostatistics* 9 (5).

Zhao, Celina. 2026. "OpenAI Breakthrough Triggers 'Existential Crisis' in Math." *Science*, ahead of print, September. <https://doi.org/10.1126/science.zbgmw3q>.

Resources

- [Logistic Regression 3-class Classifier](#) from `sklearn`

Next lecture

- Negative log-likelihood, geometric interpretation, implementation

Appendix

One-vs-rest classifier

```
In [41]: # Load the Iris dataset
iris = load_iris()
iris_X, iris_y = iris.data, iris.target

# Binarize the output
iris_y_binarized = label_binarize(iris_y, classes=[0, 1, 2])

# Split the dataset into training and testing sets
iris_X_train, iris_X_test, iris_y_train, iris_y_test = train_test_split(
    iris_X,
    iris_y_binarized,
    test_size=0.2,
    random_state=42,
    stratify=iris_y
)
```

One-vs-rest classifier

```
In [42]: # Train one binary classifier for each class

iris_classifiers = []
for class_index in range(len(iris.target_names)):
    iris_classifier = LogisticRegression(max_iter=1000)
    iris_classifier.fit(iris_X_train, iris_y_train[:, class_index])
    iris_classifiers.append(iris_classifier)
```

Each logistic regression finds a **hyperplane** in a **4-dimensional space** that separates the data into two classes.

One-vs-rest classifier

```
In [43]: # Predict on a new sample
new_sample = iris_X_test[0].reshape(1, -1)
confidence_scores = np.array([
    classifier.decision_function(new_sample).item()
    for classifier in iris_classifiers
])

# Final assignment
final_class = np.argmax(confidence_scores)

# Printing the result
print(f"Final class assigned: {iris.target_names[final_class]}")
print(f"True class: {iris.target_names[np.argmax(iris_y_test[0])]}")
```

Final class assigned: setosa
True class: setosa

label_binarize

```
In [44]: # Original class labels
example_labels = np.array([0, 1, 2, 0, 1, 2, 1, 0])

# Binarize the labels
example_labels_binarized = label_binarize(example_labels, classes=[0, 1, 2])

print("Binarized labels:\n", example_labels_binarized)

# Convert binarized labels back to the original numerical values
original_labels = [np.argmax(label_vector) for label_vector in example_labels_binarized]
print("Original labels:\n", original_labels)
```

Binarized labels:

```
[[1 0 0]
 [0 1 0]
 [0 0 1]
 [1 0 0]
 [0 1 0]
 [0 0 1]
 [0 1 0]
 [1 0 0]]
```

Original labels:

```
[np.int64(0), np.int64(1), np.int64(2), np.int64(0), np.int64(1), np.int64(2), np.int64(1), np.int64(0)]
```

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Attribution: "An image of a sad Chihuahua amigurumi with a birthday hat." Generated by DALL-E, via ChatGPT (GPT-4), OpenAI, September 16, 2024.